Name = Aryan
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BTech CSE - A

Assignment - 2

Ans 1 : 1) Entity Relationship Diagram
2) Relational
3) Object - Oriented
4) Integrity

Ans 2 : Data abstraction means hiding complex database details and showing only important information to users. In a healthcare system, it helps doctors, nurses and staff access patients, records easily.

There are three levels of data abstraction:

1) Physical Level : Describes how data is stored in the database. Users do not interact with this level.

2) Logical Level : Defines what data is stored, such as patient details, doctor information and appointment & the relationship between them.

3) View Level : Show only required information to different users. For example, doctors can see medical history, nurses can see patient details.

Ans 3: Benefits :-

1) Improves data security & privacy.
2) Makes the system easy to use for healthcare staff.
3) Provides data independence, so changes in database structure do not affect users.
4) Helps in better data management.

Challenges:

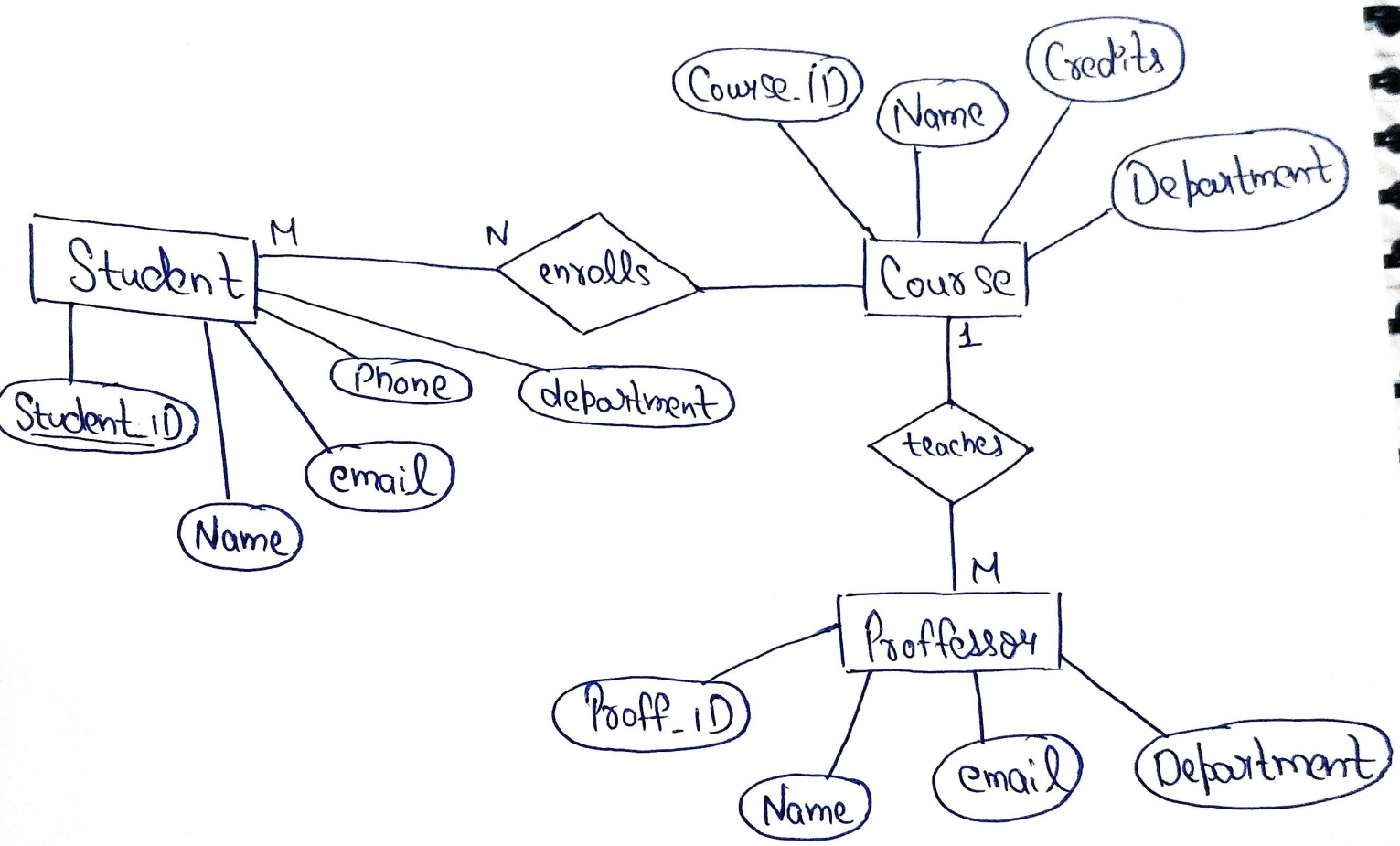
1) Implementation Complexity
2) Performance overload
3) High development cost
4) Security risks if poorly implemented
5) Training requirements.

Section C:

Student (Student_ID (PK), Name, Email, Phone, Department)

Course (Course_ID (PK), Name, Credits, Department)

Attributes (Proff_ID (PK), Name, Department)

Relationship:

1. Student - Enrolls - Course m : m
2. Proff - Teaches - Course 1 : m